

FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

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PROFESSIONAL PROFILE

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

EDUCATION

M.Sc. Cinema and Media Engineering - Politecnico di Torino **2024 - 2026**

- **Expected Grade:** 110/110 Cum Laude (Graduating Dec 2026)

B.Sc. Cinema and Media Engineering - Politecnico di Torino **2021 - 2024**

- **Final Grade:** 100/110

WORK EXPERIENCE

Lemons In The Room | Florence, Italy **03/2026 - Present**

XR Master Thesis - R&D Graphics Programmer

- Engineered a real-time HLSL Raymarching framework for Signed Distance Fields (SDFs) in Unity (URP), enabling procedural, infinite-resolution topological morphing driven by biometric data.
- Developed an in-engine authoring tool with custom Shader Graph nodes and dynamic parameter routing, allowing artists and clinicians to visually compose and map live BLE biometric streams (ECG, HRV, breathing) to SDF morphology.
- Implemented Dynamic Bounding Volumes and adaptive step-size scaling to cut GPU fill-rate costs, guaranteeing a stable 72Hz framerate on standalone Meta Quest 3 hardware.

Level Up Lab | Turin, Italy **01/2025 - Present**

Technical Artist & Gameplay Programmer

- Programmed core gameplay mechanics in C# (Unity) and designed the technical architecture for the "Line Operator" project.
- Acted as Project Manager and Technical Producer, coordinating multidisciplinary teams of designers and artists to oversee artistic vision and technical viability.

- Developed custom HLSL Shaders and post-processing pipelines to establish the game's unique visual identity.

Politecnico di Torino | Turin, Italy

10/2024 - 01/2025

Teaching Assistant (GUI)

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

RKH SRL | Turin, Italy

09/2023 - 06/2024

Video Artist

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.

SKILLS

- **Languages:** C++, C#, Python, HLSL/GLSL
- **Engines:** Unity, Unreal Engine 5, Custom OpenGL Frameworks
- **Software:** Blender, DaVinci Resolve, Jira, Git
- **Management:** Project Management, Technical Direction, Agile Product Strategy

PERSONAL PROJECTS

- **Smart Cable Generator (Blender Tool):** Engineered a Python add-on for Blender to procedurally generate parametric hanging cables across multi-object modular assets with scale-independent catenary physics and automated mesh conversion for game engines.
- **Graphics Lab (Custom C++ Engine):** Engineered a custom OpenGL 3D rendering framework from scratch. Implemented Deferred Shading, Physically Based Rendering (PBR), and Silhouette Parallax Occlusion Mapping (SPOM) to study low-level GPU pipelines.
- **UE5 Tech Art Pipeline:** Developed procedural environments using PCG. Scripted a homing projectile system in Niagara and integrated MetaHumans with custom groom collision physics.
- **Character Rigging & Tooling:** Built complete custom FK/IK character rigs (Blender), painting skin weights and setting up constraint networks. Developed Python automation scripts to streamline 3D asset export pipelines into game engines.
- **Awaken Asylum & Nihil:** Developed an immersive VR educational experience (Unity) and a 2D Metroidvania prototype emphasizing complex transformation mechanics. Managed end-to-end technical production.